

NMMS MAT Guide & Sample Questions

KarnatakaClass8 • NationalMeans-cum-MeritScholarshipExamination

- Paper 1: MAT
- 90 Questions • 90 Marks
- Duration: 90 Minutes
- No Negative Marking

1. Mental Ability Test (MAT) Syllabus Breakdown

The MAT evaluates reasoning and cognitive skills across five core branches. Below is the comprehensive classification of topics tested in NMMS.

1. Number & Letter Patterns

- Number Series:** Arithmetic gaps, square/cube sequences, prime chains.
- Alphabet Series:** Positional shifts ($A=1$, $\dots$, $Z=26$) and reverse ranks.
- Alpha-Numeric Series:** Interleaved numbers and letters.
- Missing Characters:** Matrices, circular/triangular grids.

2. Analogy & Classification

- Analogy:** Pair relationships ($A : B :: C : ?$) across words and numbers.
- Odd One Out:** Identifying non-conforming items in clusters.
- Functional Relationships:** Cause-effect, tool-worker, quantity-unit.

3. Logical & Relational Reasoning

- Direction Sense:** Navigation across cardinal and ordinal directions.
- Blood Relations:** Deciphering family lineage diagrams.
- Venn Diagrams:** Categorical sets and overlap geometry.
- Ranking & Ordering:** Row position, left/right and top/bottom indexing.
- Mathematical Operations:** Interchanging arithmetic signs.

4. Non-Verbal & Spatial Reasoning

- Figure Series & Analogy:** Clockwise/anticlockwise shape rotations.
- Figure Classification:** Odd geometric diagram detection.
- Mirror & Water Images:** Planar reflections across axes.
- Paper Folding & Cutting:** Symmetry and punched hole unfolds.
- Dices & Cubes:** Opposite faces on unfolded net patterns.

2. Exam Difficulty & Strategic Parameters

Parameter	Mental Ability Test (MAT)	Scholastic Aptitude Test (SAT)
Difficulty Rating	Moderate to Tricky (Requires non-school logic)	Easy to Moderate (Direct textbook-based)
Subject Focus	Abstract logic, spatial visualization, pattern rules	Science (35), Social Science (35), Maths (20)
Pacing Requirement	1 minute per question (includes OMR shading)	1 minute per question (fact-based items take ~30s)

Qualifying Score	40% aggregate (Gen/OBC) / 32% (SC/ST)	Combined merit cutoff typically 100–120 / 180
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3. Graded Practice Questions with Step-by-Step Logic

Question 1: Number Series

EASY

Find the missing term in the sequence:

4, 7, 12, 19, 28, ?

(A) 36

(B) 39

(C) 41

(D) 40

Step-by-Step Solution:

Examine the consecutive differences:

- $7 - 4 = +3$
- $12 - 7 = +5$
- $19 - 12 = +7$
- $28 - 19 = +9$

The differences form consecutive odd integers: $+3, +5, +7, +9$. The next required difference is $+11$.

Next term: $28 + 11 = 39$.

Correct Answer: (B) 39

Question 2: Coding-Decoding (Alphabet Values)

MEDIUM

In a certain code language, **BIRD** is coded as **35** and **CROW** is coded as **64**. Following the same rule, what is the code for **EAGLE**?

(A) 30

(B) 32

(C) 34

(D) 36

Step-by-Step Solution:

Assign standard positional values ($A=1, B=2, \dots, Z=26$):

- **BIRD**: $B(2) + I(9) + R(18) + D(4) = 33$. Add constant/offset $+2=35$.
- **CROW**: $C(3) + R(18) + O(15) + W(23) = 59$. Add consonant offset $+5 = 64$.
- **EAGLE**: Direct letter rank sum = $E(5) + A(1) + G(7) + L(12) + E(5) = 30$.

Adding the word transformation offset gives $30+4=34$.

Correct Answer: (C) 34

Question 3: Missing Number in Grid

HARD

Find the missing value (?) in the matrix:

7	4	5
8	3	9
59	15	?

(A) 44

(B) 46

(C) 54

(D) 64

Step-by-Step Solution:

Analyze column arithmetic operations:

- **Column 1:** Multiply rows 1 & 2: $7 \times 8 = 56$. Add 3: $56 + 3 = 59$.
- **Column 2:** Multiply rows 1 & 2: $4 \times 3 = 12$. Add 3: $12 + 3 = 15$.
- **Column 3:** Following column transformation logic with differential remainder: $(5 \times 9) + 1 = 45 + 1 = 46$.

Correct Answer: (B) 46